

Motion Synergy API Release Notes

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1 Motion Synergy API Release Notes

This document describes all significant changes in the current release and provides a record of changes for past releases

1.1 CURRENT - 24th Dec 2025

1.1.1 New Features and Enhancements

- Initialization can now be interrupted using an 'Emergency Stop All' from the GUI.
- Biquad filters have been moved into the Motion Tuning window, available from the View and Edit Parameters.
- Tuning parameters can now be updated by connecting to an axis without needing to complete initialization.

1.1.2 Bug Fixes

- DCLD-900: Position/CurrentLoopInputsInUse config field in MotionTuningConfigs always false.
- DCLD-913: Show error message tooltips in travel length combobox on Reconfigure Window.
- DCLD-928: UI will exit when trying to connect an individual axis after 'Initialize and Home All' fails.
- Position loop values set to zero if the user clicks 'Write to NVRAM' from the Motion Tuning screen without making any changes.

1.1.3 Known Limitations

- None

1.1.4 Known Defects

- DCLD-889: Sometimes IXAAT CAN Drivers are not detected when they are already installed
- Various minor UI bugs

1.2 4.0.16022 - 18th Dec 2025

1.2.1 New Features and Enhancements

- Emulated axes now use semi-representative trace capture data files so trace capture and motion tuning functions can be demonstrated in emulation.
- Bode plot data capture can now be performed on Linux systems via the Python example application.
- Support loading the MotionSynergyAPI as a DLL from C++ applications. The C++ example console application is now distributed in the standard release.
- Confirmed compatibility with latest CMake (v4.2.1)

- Auto-connect feature disables the baud rate and address drop down boxes.
- Renamed 'Reconfigure' button to 'Reconfigure Axes', added a confirmation message, and added a step to backup the current configuration prior to modification.
- Emulated axes are now indicated by '(Emulated)' in the tab header on the GUI.
- Refreshed screenshots in User Manual.

1.2.2 Bug Fixes

- None

1.2.3 Known Limitations

- None

1.2.4 Known Defects

- DCLD-889: Sometimes IXAAT CAN Drivers are not detected when they are already installed
- DCLD-928: UI will exit when trying to connect an individual axis after 'Initialize and Home All' fails.
- Various minor UI bugs

1.3 4.0.15921 - 26th Nov 2025

1.3.1 New Features and Enhancements

- Users are now notified when screenshots are taken during Bode Plot Analysis.
- Multi-Position Bode plot results are now saved.
- DotNet libraries available for .NET 4.62 (Windows-only) and .NET 8.0 (Windows and Linux).
- C++ source code is now C++20 compliant (requires a recent version of Visual Studio to build).
- C++ source code will now build on Ubuntu 24.04.
- External C++ libraries (e.g. boost) managed using vcpkg package manager.
- Updated Packaged Docs to latest version.
- Boost version updated to 1.89.
- Updated various libraries including Lua (now 5.4.8), LuaUnit and others.
- Tuning Files are now removed when product is reconfigured
- Updated terminology in the tuning page and added finer grain apply options
- Position Error is now available through the API
- Users can now setup a sine wave user defined profile through Lua interface.

1.3.2 Bug Fixes

- DCLD-878: Cannot Do NVRAM Action during a drive error
- DCLD-653: Get Phase Angle Button Disappears
- DCLD-602: After saving to NVRAM during motion tuning, the device may behave in an erratic manner. Users are recommended to re-initialize after writing to NVRAM during motion tuning
- DCLD-800: Writing to NVRAM GUI occasionally crashes after performing a write.
- DCLD-564: Timeout Error during Homing
- DCLD-770: Points are now properly interpolated in bode plots
- DCLD-864: Heading Tabs Disappear when vertical scrolling in main window

1.3.3 Known Limitations

1.3.4 Known Defects

- DCLD-889: Sometimes IXAAT CAN Drivers are not detected when they are already installed
- Various minor UI bugs

1.4 3.7.15620 - 16th Oct 2025

1.4.1 New Features and Enhancements

- Added support for DOF-10 Product
- Implemented Bode Plot Analysis at multiple positions along a single axis
- Data of previous runs of Bode Plots will now be saved
- Sine Sweeps can now be canceled with an E-Stop
- Refactored Main Screen
- Refactored GUI for Axis Manipulation
- Will now notify users when the calculated current loop parameters are changed

1.4.2 Bug Fixes

- DCLD-760: RMS Calculation should not be visible on spectrum plots.
- DCLD-801: An unrecoverable error is thrown when attempting to move towards a limit that has already been reached. This is meant to be recoverable.
- DCLD-809: Users in regions with comma delimited decimals may experience crashing if their system is not set to USA units.

1.4.3 Known Limitations

- Plotting with emulated hardware is currently not supported.

1.4.4 Known Defects

- DCLD-602: After saving to NVRAM during motion tuning, the device may behave in an erratic manner. Users are recommended to re-initialize after writing to NVRAM during motion tuning
- DCLD-800: Writing to NVRAM GUI occasionally crashes after performing a write.
- DCLD-580: Motion Tuning GUI sometimes allows the user to perform a move when in an unknown state causing crashing.
- DCLD-783: Incorrect commanded position is shown in GUI during coordinated motion while in UDP Follower mode
- DCLD-508: RaspberryPi architecture not recognized when building the Console Application
- DCLD-683: Libraries occasionally fail to load
- Various minor UI bugs

1.5 3.7.15434 - 28th Aug 2025

1.5.1 New Features and Enhancements

- Various updates and upgrades to the Bode Plot output visualization.
- Updated documentation to reflect known bugs.

1.5.2 Bug Fixes

- DCLD-660: Fixed bug where Linux Release builds would fail to commutate during initialization. Users should now be able to properly initialize their devices on a Linux Release build.

1.5.3 Known Limitations

- Plotting with emulated hardware is currently not supported.

1.5.4 Known Defects

- DCLD-602: After saving to NVRAM during motion tuning, the device may behave in an erratic manner. Users are recommended to re-initialize after writing to NVRAM during motion tuning
- DCLD-800: Writing to NVRAM GUI occasionally crashes after performing a write.
- DCLD-801: An unrecoverable error is thrown when attempting to move towards a limit that has already been reached. This is meant to be recoverable.
- DCLD-580: Motion Tuning GUI sometimes allows the user to perform a move when in an unknown state causing crashing.
- DCLD-760: RMS Calculation should not be visible on spectrum plots.
- DCLD-783: Incorrect commanded position is shown in GUI during coordinated motion while in UDP Follower mode
- DCLD-508: RaspberryPi architecture not recognized when building the Console Application

- DCLD-683: Libraries occasionally fail to load
- DCLD-809: Users in regions with comma delimited decimals may experience crashing if their system is not set to USA units.
- Various minor UI bugs

1.6 3.7.15390 29th Jul 2025

1.6.1 New Features and Enhancements

- Motion tuning feature now supports auto-detect of payload mass and damping, and subsequent calculation of velocity and acceleration feed forward parameters.
- Motion tuning feature now displays a spectrum analysis of Position Error at Rest and Move Optimization Plots, aiding in the identification of resonant frequencies.

1.6.2 Bug Fixes

- DCLD-401: UDP StartValue and StartIndex greater than 0 causes RuntimeError.
- DCLD-639: Allow Shutdown and Re-initialization of an axis in emulation (this matches the real device behavior).
- DCLD-641: Improved support for writing stepper motor configuration to NVRAM.
- DCLD-740: Adjust analog calibration for the Open Frame Stage.

1.6.3 Known Limitations

- None

1.6.4 Known Defects

- None

1.7 3.7.15172 22th May 2025

1.7.1 New Features and Enhancements

- Phase angle now accessible from Lua. See `Axis.MotionControllerConfiguration.PhaseAngleDeg` & `Axis.MotionControllerConfiguration.PhaseAngleRaw`.

1.7.2 Bug Fixes

- DCLD-709: Coordinated motion conflict with TOP - Driven axis does not respond to master axis after it's performed a TOP move. This fix also requires a firmware update from PMD.
- DCLD-638: DMCM stepper motor fails first move. Ensure the correct operating mode is selected for stepper motors.

- DCLD-636: When exiting electronic gearing mode (i.e. external control), set gear ratio and velocity to zero to ensure there is no spurious motion.

1.7.3 Known Limitations

- None

1.7.4 Known Defects

- DCLD-401: UDP StartValue and StartIndex greater than 0 causes RuntimeError

1.8 3.7.15126 8th April 2025

1.8.1 New Features and Enhancements

- Support 'On-the-fly' Updates to UDP profiles.

1.8.2 Bug Fixes

- None

1.8.3 Known Limitations

- None

1.8.4 Known Defects

- DCLD-401: UDP StartValue and StartIndex greater than 0 causes RuntimeError

1.9 3.7.15107 7th March 2025

1.9.1 New Features and Enhancements

- Support Coordinated Motion using UDP Master/Driven profiles.
- Rename the 'Invetech' namespace to 'DoverMotion'. Customers using the code from this namespace must rename it before re-building their application.
- Remove Invetech from End User Agreement.

1.9.2 Bug Fixes

- DCLD-608: Next move after disabling external control generates a spurious position error.
- DCLD-610: Cancelling auto connect closes the Discovery windows but doesn't cancel the auto-connect procedure which continues to run in the background.
- DCLD-635: Unable to find axis travel length unhandled exception generated when an axis travel length in configuration doesn't match a known product travel length.
- DCLD-642: Race condition in single line log viewer causes exception in rare circumstances.

1.9.3 Known Limitations

- None

1.9.4 Known Defects

- DCLD-401: UDP StartValue and StartIndex greater than 0 causes RuntimeError

1.103.6.12457 05 June 2024

1.10.1 New Features and Enhancements

- Allow SetCommutationParameter to be written to NVRAM. Note that only parameter 0 is allowed when reading and writing NVRAM.
- Add methods to set phase angle, and get/set commutation parameter 1.

1.10.2 Bug Fixes

- When the de-tuning dialog box appears, asking whether to de-tune or use existing NVRAM parameters, closing the dialog box now selects the NVRAM option.

1.10.3 Known Limitations

- None

1.10.4 Known Defects

- DCLD-401: UDP StartValue and StartIndex greater than 0 causes RuntimeError

1.113.6.12428 08 May 2024

1.11.1 New Features and Enhancements

- Add a new product definition - SmartStage Open Frame 150mm x 150mm.
- Update product definitions to include information required for automatic tuning.
- Add a "Tuning Window" button under the Motion Tuning advanced tab. When pressed, a new window with the following features will be created:
 - Option to apply appropriate position loop and current loop parameters to an axis, based on user-selectable bandwidth and payload.
 - Option to apply the parameters to the buffer memory, or to write them to NVRAM.
 - Option to import a tuning parameter configuration file, then apply the parameters.
 - Option to move relative a short distance, then automatically plot *position error* and *move velocity* values. The plots will also be annotated with the following: Trajectory Complete, Axis Settled, Settle Window.
 - Option to plot *position error* at rest. Two seconds of data will be captured.

- In-window Motion Complete Mode and Move profile settings.
- In-window Move absolute action.
- Move manual adjustment of position loop and current loop tuning parameters to the tuning window.
- MotionSynergyGUI now displays and logs an unhandled exception prior to exiting.
- Add interactive prompts, allowing the user to de-tune an axis, when an axis is initialized for the first time (via Initialize & Home All button).

1.11.2 Bug Fixes

- None

1.11.3 Known Limitations

- None

1.11.4 Known Defects

- DCLD-401: UDP StartValue and StartIndex greater than 0 causes RuntimeError

1.123.6.12191 01 March 2024

1.12.1 New Features and Enhancements

- Added support for SmartStage Monolithic, SmartStage Open Frame, SmartStage Shuttle and SmartStage Z-50 products.
- Biquad Filter 2 only takes effect if Biquad Filter 1 is enabled, so don't allow the user to set filter 2 if filter 1 is off.
- Various SignalStatus register bits can now be included in a trace capture.
- Attempt to retrieve and display trace capture data, regardless of whether the move completed successfully or not.
- Support an optional delay after a move completes and before a trace capture is stopped.
- Support connecting / disconnecting individual axes through the GUI to allow configuration of individual axis communications settings, etc.
- Renamed button *Reset Position Error* to *Reset Position Error & Re-energize* and button *Install* to *Install Driver*.
- Allow axis names to include the underscore (_) character.
- Simplified the UI for updating axis communications settings.
- Current position and axis flag status is now displayed by the Motion Synergy GUI after connect (previously only displayed after initialization).
- Motion Controller firmware version and phase angle now displayed on the Setup tab.

- Discover the communications settings in use by an axis by attempting to connect using different combinations of settings (i.e. baud rate, address).

1.12.2 Bug Fixes

- DCLD-481: No error reported on PMD controller chip reset. Now detects reset and reports a general axis error.
- DCLD-502: Displayed current loop values wrong for 3-phase motors (e.g. SmartStage XY).
- DCLD-507: Newly created move profile could not be selected after it was first created.
- DCLD-509: Do not set the action to take on hitting a limit switch to None during initialization. Now uses action specified in NVRAM.

1.12.3 Known Limitations

- None.

1.12.4 Known Defects

- DCLD-401: UDP StartValue and StartIndex greater than 0 causes RuntimeError

1.133.6.12025 21 December 2023

1.13.1 New Features and Enhancements

- When the action on motion error is configured to None, a simple warning is logged at the driver level but the move will appear to complete successfully.
- Created a list of commands that should not be written to NVRAM and prevent any of these commands being written to NVRAM.
- Updated the Motion Tracking help documentation and associated diagram to clarify behavior.
- Added a trace profile for plotting TOP (Trigger on Position) Gate and Trigger signals.
- Reporting of duplicate error messages in the Motion Synergy GUI has been reduced and axis alert descriptions now include the error source (i.e. axis name).
- If a connection to an axis is unsuccessful, the Motion Synergy GUI provides a prompt and button to emulate the axis.
- A new MoveRelativeToFlagOpenLoop(flag_name, offset) command has been added which can be used to move a SmartStage axis to a flag prior to calling Initialize().
- It's now possible to plot Trigger On Position (TOP) trigger and gate signals in a trace capture.
- Improved wording and button states for starting, retrieving and stopping trace capture data.
- The Motion Synergy GUI now polls for position and flag status updates, removing the need to manually click the Refresh button.
- Improved wording and diagram for the Motion Tracking context sensitive help.
- Updated biquad filter defaults and error checking in the UI to make it more intuitive.

- The SmartStage axis initialize script now checks if an axis is on a positive or negative limit prior to initialization and if so, moves it away from the limit using a `MoveRelativeToFlagOpenLoop()` command before calling `Initialize`.
- A warning is generated when attempting to perform `Initialize` on a positive or negative limit. This warning indicates commutation may fail in this state due to a collision with a hard stop.
- Trace profiles are no longer listed in `MotionSynergyGUI.cfg`. Instead they are discovered automatically from `Instrument.cfg` using the suffix `'TraceProfile'`.
- The `Initialize` and `Home` buttons are now combined into a single button on the individual axes tabs. This now executes the `Initialize` and `Home` script which is more consistent with the `Initialize` and `Home All` command.
- The configuration parameter `MaxMotorCurrentPercent` is now applicable for both algorithmic and direct set commutation. Default is 25% and can be increased up to a maximum of 50% for stages with heavy payloads.
- On startup of the Motion Synergy GUI, detect missing Microsoft Visual C++ Runtime Libraries and prompt the user to install them.

1.13.2 Bug Fixes

- DCLD-464: `CppConsoleApplication` reports spurious `Varnum: Value 10005 is not a valid value` error when a connection to hardware cannot be established.
- DCLD-466: Unknown PMD commands cause reading from NVRAM to fail. Updated to report a warning and continue with the read.
- DCLD-476: Race condition in `AlertCapture` class fixed. Showed up in a unit test.
- DCLD-480: MotionSynergy GUI can crash on connection lost.

1.13.3 Known Limitations

- None.

1.13.4 Known Defects

- DCLD-401: UDP `StartValue` and `StartIndex` greater than 0 causes `RuntimeError`

1.14v3.5.11906 10 November 2023

1.14.1 New Features and Enhancements

- This is a major update which adds support for Linux (available on request), specifically x64 Ubuntu 22.04 and arm64 Ubuntu 22.04 (e.g. Raspberry Pi).
- Python support is no longer tied to a specific embedded version. Python.Net can be 'pip' installed with Python 3.8 or above.
- The examples are now targetted for Visual Studio 2022 on Windows. Visual Studio 2019 support is no longer available.

- Updated .NET Framework version dependency from 4.6.1 to 4.6.2 for Windows and .NET 6.0 for Linux.
- Simplified IMotionSynergyAPI interface by rolling IMotionSynergyAPIBase functionality into IMotionSynergyAPI.
- Added support for continuous moves with a new MoveContinuous() command.
- Added new tab to the UI which can be used to control an XY SmartStage and optional DOF-5 using the mouse.
- Updated the example C++ console application to include support for an XYZ setup.

1.14.2 Bug Fixes

- DCLD-421: MotionSynergyAPI does not run on older versions of Windows 10.
- DCLD-422: MotionSynergy depends on VC++ Runtime library but doesn't automatically install it.
- DCLD-449: SetMotorCommand for algorithmic mode is hard coded, and should be using the instrument.cfg value.

1.14.3 Known Limitations

- None.

1.14.4 Known Defects

- None.

1.15v3.4.11658 13 July 2023

1.15.1 New Features and Enhancements

- Enhanced First Time Startup experience:
 - Add multiple products.
 - Rename axes.
 - Configure axis communication address.
 - Option to install CAN and MOXA drivers as required.
- Add stepper motor support to the DMCM.
- Added Tuning functions to the Advanced features which include:
 - Set Position Loop parameters.
 - Set Current Loop parameters.
 - Persist above settings to NVRAM.
- Plotting of trace capture data within the MotionSynergyGUI.
- Allow NVRAM settings to be updated without the need to Initialize each axis.

- Added the following new methods to the axis sub-interface IMotionControllerConfiguration:
 - Get/SetMotionTuning()

1.15.2 Bug Fixes

- None.

1.15.3 Known Limitations

- None.

1.15.4 Known Defects

- None.

1.16v3.4.11014 23 March 2023

1.16.1 New Features and Enhancements

- Enhanced First Time Startup experience:
 - Added SmartStage Linear to the available products.
 - Added ability to select the Travel Length for the SmartStage XY and SmartStage Linear products.
- Branding Updates:
 - Renamed EngineeringGUI to MotionSynergyGUI
 - Updated Application icons and Installer to use the MotionSynergy logo.
- Added Setup functions to the Advanced features which include:
 - Set Motion Tracking parameters including motion complete mode, settle window and settle time.
 - Set Motion Error parameters including position error limit and action on motion error.
 - Persist above settings to NVRAM.
- Removed IAxis methods:
 - GetLifetimeMoveCount()
 - GetLifetimeDistanceTravelled()
- Moved the following IAxis methods to a new sub-interface IMotionControllerConfiguration:
 - Get/SetPositionErrorMargin() -> Get/SetMotionError()
 - RestoreMotionControllerConfiguration() -> Restore()
 - SaveMotionControllerConfiguration() -> Save()
 - RestoreMotionControllerConfigurationToNVRAM() -> RestoreToNVRAM()
- Added the following new methods to the axis sub-interface IMotionControllerConfiguration:

- Get/SetMotionTracking()
 - Get/SetMotionError()
 - WriteToNVRAM()
- Remove the following methods from the Lua scripting interface - these are not needed when writing workflows:
 - RestoreMotionControllerConfiguration()
 - SaveMotionControllerConfiguration()
 - RestoreMotionControllerConfigurationToNVRAM()

1.16.2 Bug Fixes

- DCLD-278: Ensure GUI populates field with a valid distance and direction when using the MoveRelativeToFlag() command.
- DCLD-285: Python Console Example Initialize Bug
- DCLD-291: Intermittent failure to connect over RS232 during initialization. Once the failure occurs a power cycle of the hardware was required.

1.16.3 Known Limitations

- Control of a stepper motor through the DMCM is not supported.

1.16.4 Known Defects

- None.
-

1.17v3.3.10882 21 November 2022

1.17.1 New Features and Enhancements

- Support an 'advanced commutation mode' which provides more robust commutation for SmartStage XY products.
- Trace start/stop can now be triggered manually in addition to on motion start / stop.
- Magellan Controller configuration stored in NVRAM can be saved to a Magellan Controller script file.
- Magellan Controller configuration can be restored from a Magellan Controller script file to working memory or persisted to NVRAM.
- The Advanced Features expanding panel in the Engineering GUI now expands horizontally to make better use of screen real estate.
- The two general purpose inputs provided by each motor controller can be read using a digital input device.

1.17.2 Bug Fixes

- DCLD-257: Rapid polling of GetActualPosition can cause mutex destroyed while busy exception.
- DCLD-264: Race condition when error raised during script completion, causing invalid state transition exception.

1.17.3 Known Limitations

- Control of a stepper motor through the DMCM is not supported.

1.17.4 Known Defects

- None.
-

1.18v3.3.10632 06 October 2022

1.18.1 New Features and Enhancements

- Software releases are now packaged as a WinRAR self-extracting archive (.exe) which requires acceptance of the license terms before proceeding.
- Simplified axis interface for returning positions. Now have GetCommandedPosition() and GetActualPosition() which replace GetCurrentPosition(), GetEncoderPosition() and GetTargetPosition().
- Axis moves no longer wait for the 'axis settled' status bit to be set. Instead, only the 'move complete' status bit is checked.
- Axis properties can now be retrieved asynchronously during a move.
- New 'MotorCurrent' property added which returns the RMS current being drawn by the motor at a given instant (albeit with some latency due to communications).

1.18.2 Bug Fixes

- DCLD-160: Temp/Current Tracking Error reports a general error rather than a specific error.

1.18.3 Known Limitations

- Control of a stepper motor through the DMCM is not supported.

1.18.4 Known Defects

- None.
-

1.19v3.3.10488 24 August 2022

1.19.1 New Features and Enhancements

- Added an example LabVIEW project demonstrating how to access the *Motion Synergy API*.
- Magellan Motion Control documentation is now included as part of the release package.
- When running CMake to create the C++ example solution, a sensible error message is now displayed if the EngineeringGUI first time startup has not been run.
- Documentation is automatically converted to pdf format and includes a date-based version in the filename.
- Added a section to the User Manual on configuring multiple communication links.
- Added a section to the User Manual documenting how to change the amount of information logged and how to configure logging in general.
- Added a setting to the EngineeringGUI.cfg to control whether 'Initialize and Home All' is run on startup. Default is not to run.
- Added a range check on the TOP pulse width to restrict it to 50 to 204,800 ns, with an increment of 50. Documented these limitations as well as the limitation it is not supported by firmware versions <= 1.2. Changed the pulse width data type from a double to an int.
- Added an argument to Initialize() to specify whether the Initialize script is run as part of initialization.
- Updated the C# console application to not run the Initialize script during initialization.
- Updated the C# examples code to match the C++ code.
- Changed the default logging level from *Debug* (Severity 1) to *Info* (Severity 2) to reduce the amount of data written to the logs.
- Updated the createVisualStudioSolution.ps1 to report an error if the generated paths will be too long.
- Reduced standard profile max velocity from 150mm/s to 100mm/s.
- Updated configurations to use serial communications by default.
- Clarified axis Stop / EStop behavior.
- Added support for manually setting the general purpose output (AxisOut1) (see SetDigitalOutputTrigger).
- Added support for configuring a general purpose output (AxisOut1) to signal various axis motion state changes (see SetDigitalOutputTrigger).

1.19.2 Bug Fixes

- EngineeringGUI: Do not enable axis commands in the case where initialization fails or when an individual axis command fails.

1.19.3 Known Limitations

- Control of a stepper motor through the DMCM is not supported.

1.19.4 Known Defects

- DCLD-160: Temp/Current Tracking Error reports a general error rather than a specific error.
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1.20.1 New Features and Enhancements

- Added an *Initialize()* method to *IMotionSynergyAPIBase* which doesn't require a progress parameter to be supplied.
- Added a boolean flag to *InstrumentSettingsBase* called *ConvertRelativePathsToAbsolute* which will cause relative paths to be converted to absolute paths prior to their use. This allows an application to have a current directory which is different from the location of the Motion Synergy API dlls.

1.20.2 Bug Fixes

- None

1.20.3 Known Limitations

- Control of a stepper motor through the DMCM is not supported.
- First time startup must be run from the EngineeringGUI before building the C++ example or a CMake error will be reported.

1.20.4 Known Defects

- DCLD-160: Temp/Current Tracking Error reports a general error rather than a specific error.